

Lab 03: IPv4 Subnetting and Address Planning

Duration: 1 hour | In-Lab Assessment

Tools: Cisco Packet Tracer

Objectives

By the end of this lab, students will be able to:

- Understand the basics of IPv4 subnetting and CIDR notation.
- Subnet a fixed-length network to meet given host requirements.
- Apply Variable Length Subnet Mask (VLSM) to allocate address blocks efficiently.
- Design and implement a multi-LAN topology in Cisco Packet Tracer using VLSM.
- Configure classless routing using RIP version 2 and explain the purpose of disabling automatic summarization.
- Verify end-to-end connectivity and inspect ARP tables on hosts.

Lab Description

Part A: Basic IPv4 Subnetting

1. Theory Review:

- Recall that a subnet mask of $/n$ means the first n bits are network/subnet and the remaining $32 - n$ bits host.
- Compute # of subnets: 2^{Δ} , where Δ is the number of bits borrowed.
- Compute hosts per subnet: $2^{\text{host bits}} - 2$, where the first address is the network address and the last address is the broadcast address.

2. Exercise: Given network 172.16.0.0/16, create four equal-sized subnets.

- Determine how many bits to borrow for 4 subnets.
- Write each subnet's network address, subnet mask (in dotted decimal and CIDR), usable host range, and broadcast address.

Part B: Variable Length Subnet Mask (VLSM) Design

1. Host Requirements Table:

Subnet	Required Hosts
netA	14
netB	28
netC	2
netD	7
netE	28

2. Design Steps:

- (a) Order subnets by descending host requirement.
- (b) For each, choose the smallest CIDR block that accommodates the required hosts.
- (c) Allocate subnets sequentially from 204.15.5.0/24, incrementing network address accordingly.

3. Resulting VLSM Plan:

Subnet	Network	Mask	Usable Hosts Range
netB	204.15.5.0	/27(255.255.255.224)	204.15.5.1 – 204.15.5.30
netE	204.15.5.32	/27	204.15.5.33 – 204.15.5.62
netA	204.15.5.64	/28(255.255.255.240)	204.15.5.65 – 204.15.5.78
netD	204.15.5.80	/28	204.15.5.81 – 204.15.5.94
netC	204.15.5.96	/30(255.255.255.252)	204.15.5.97 – 204.15.5.98

Note: The last address in each subnet is reserved as the broadcast address and is not usable for host assignment.

Part C: RIP Configuration and Classless Routing

To enable routing between the subnets defined above, configure RIP version 2 with the following guidelines:

1. On each router, enter configuration mode and enable RIP:

```
router rip
  version 2
  no auto-summary
  network 204.15.5.0
```

2. Why RIP Version 2?

RIP version 1 only supports classful routing and does not advertise subnet masks. Since this lab uses VLSM (variable-length subnet masks), it is essential to use RIP version 2, which supports classless routing and includes subnet mask information in its updates.

3. Why no auto-summary?

By default, RIP summarizes routes to their classful boundaries (e.g., summarizing all subnets under 204.15.5.0/24). This behavior can cause incorrect routing in discontinuous networks. Disabling automatic summarization ensures that each subnet is advertised with its correct subnet mask, allowing precise routing between non-contiguous subnets.

4. Verify RIP routing with the following commands:

- `show ip protocols` – to confirm RIP v2 is active.
- `show ip route` – to verify RIP routes are installed.
- `debug ip rip` – to monitor RIP updates in real-time (optional).